AOUANOUK Slimane

+33 7 83 54 30 26 — aouanoukslim@gmail.com — LinkedIn — slimaneaouanouk.com
Mechanical engineering student at Arts et Métiers specializing in robotics and AI.

Arts et Métiers engineering student specializing in robotics and AI, with research experience in **trajectory optimization, robot perception, real-time control and physical validation**. **First author of a robotics manuscript currently under journal review**. Currently developing a **Vision-Language-Action pipeline for assistive humanoid manipulation**.

EDUCATION — Combined GPA : 3.92 / 4.00

Arts et Métiers Institute of Technology (ENSAM) — Paris, France **Sept 2024 – Jul 2027**
Combined BS × MS Engineering Degree **GPA: 3.86/4.00**

- **Ranked top 12% of cohort** — 142 out of 1,223 students.
- **Mechanical engineering foundation:** multidisciplinary training in mechanical design, solid mechanics, numerical optimization, manufacturing and experimental validation.
- **Selected for the final-year Robotics specialization (2026–27)**, with coursework at both Arts et Métiers and Sorbonne Université.
- **Advanced robotics studies** connect robot perception, sensor fusion and state estimation — including Bayesian methods and Kalman filtering — with machine learning and control for autonomous robotic systems.

Lycée Raspail — Paris, France **Sept 2022 – Jul 2024**
Intensive Preparatory Classes (CPGE), PTSI / PT* **GPA: 4.00 / 4.00**
Intensive two-year postsecondary program preparing students for competitive entrance examinations to French engineering schools, with **35+ scheduled class hours per week plus oral examinations**.
Relevant Coursework: Advanced Mathematics, Physics, Mechanical Engineering and Engineering Sciences.

WORK EXPERIENCE

Trajectory Optimization and Collision Avoidance for Autonomous Systems **Jan–Aug 2026**
Robotics Research Project & Internship — PIMM Lab / CNRS@CREATE; Paris (Jan–May) → Singapore (Jun–Aug) — Supervisors: Prof. Amine Ammar, Prof. Francisco Chinesta and Prof. Dominique Baillargeat

- **Co-developed a Python/OpenCV visual-localization pipeline** using a Basler overhead camera and ArUco markers to estimate rover position and heading for closed-loop mobile-robot navigation.
- **Calibrated camera intrinsics using 20 ChArUco views**, achieving **0.191 px reprojection RMS**; combined lens-distortion correction with metric homography to reduce mean localization error **84%, from 36.3 to 5.8 mm**, on an independent nine-point ground grid.
- **Diagnosed and corrected height-induced parallax**, reducing mean rover-marker localization error from **60.8 to 11.6 mm** across five physical validation positions.
- **Profiled and optimized vision processing through image downsampling and point-wise distortion correction**, integrating localization, trajectory correction, serial rover control, and experiment logging at a median **19.84–19.86 Hz**.
- **Validated the integrated navigation stack through two frozen 20-run physical campaigns**, achieving **40/40 goal arrivals with zero collisions** and collecting synchronized control and trajectory logs.
- **Evaluated trajectory-planning and control configurations through a 100,800-case simulation campaign** and quantified simulation-to-real discrepancies across **39 paired physical and digital-twin runs**.
- **Co-developed pytest unit and regression tests** covering calibration, homography, controller behavior, and analytical gradients, including finite-difference checks for numerical consistency.
- **First author of a robotics manuscript currently under journal review**, presenting the trajectory-planning framework, experimental validation, and simulation-to-real evaluation.

SELECTED PROJECTS

Humanoid Robot Learning for Assistive Manipulation — VLA **2026–27**
Sorbonne Université · Supervised by Prof. Olivier Bruneau, LURPA / ENS Paris-Saclay

- **Developing an end-to-end Vision-Language-Action (VLA) pipeline** for autonomous manipulation on a Unitree G1 humanoid, from teleoperated demonstrations to real-world deployment.
- **Targeting OpenPI (π_0/$\pi_{0.5}$) policy fine-tuning** using imitation learning and multimodal datasets combining natural-language instructions, RGB/RGB-D observations, and robot actions.
- **Exploring language-conditioned manipulation policies** for household assistance tasks, targeting robust execution, generalization to unseen configurations, and quantitative real-world evaluation.
- **Exploring AI-based motion optimization in MuJoCo or NVIDIA Isaac Lab (Isaac Sim)**, with motion capture and human-to-robot retargeting envisaged to adapt human movements to the robot.

SELECTED PROJECTS (CONTINUED)

CNAM Bicopter Control **Jan 2026**
Arts et Métiers / CNAM — Control Systems Laboratory
- **Co-identified a second-order elevation model** from experimental bicopter response data and formulated a discrete state-space representation in MATLAB/Simulink.
- **Co-designed state feedback with integral action using pole placement**, simulated the controller and implemented it on dSPACE at a **50 ms sampling period**.

Aircraft Brake System Design Project **2025**
Arts et Métiers — Individual Preliminary Mechanical Design
- Designed a complete **mechanical braking system** for a light aircraft wheel composed of **10+ components** using **Fusion 360**; built a physics-based functional architecture ensuring torque transmission across 5+ mechanical interfaces; quantitatively modeled load paths and mechanical constraints; performed engineering validations including **bearing preload analysis**, **bolt slip criteria**, **shaft stress verification**, and **brake disc thermal dissipation**; ensured system consistency through analytical checks and safety margins

TIPE — Experimental Impact Testing & Python Video Analysis **2022–24**
Individual CPGE Experimental Project
- **Designed and built a 46 cm pendulum-style impact rig** with four interchangeable contact geometries: 6, 12 and 24 mm cylinders and a flat plate, using controlled ball positioning and comparable impact conditions.
- **Processed impact videos in Python to track the ball and impactor**; used NumPy, SciPy and Matplotlib to analyze position data, fit pre- and post-impact linear regressions, estimate velocities and display uncertainty bars.
- **Measured a 4.0 m/s ball exit velocity from a 2.2 m/s incoming impactor** in a documented **12 mm-cylinder test at 3 psi**, corresponding to a ball-to-impactor speed ratio of approximately **1.8**.

Foundry, Machining & Metrology **2025**
Arts et Métiers — Manufacturing and Experimental Inspection
- **Made the sand mold and cast an AlSi7 bearing support**, defining the parting plane, core positioning, draft angles and machining allowances.
- Prepared **2 machining phase sheets** and machined the casting on a conventional lathe and CNC mill, setting workholding, alignment and workpiece datums.
- Inspected **6 dimensional and geometric characteristics** using a CMM; measured **0.015 mm flatness against a 0.100 mm limit** and **0.018 mm cylindricity against 0.020 mm**.
- Identified insufficient machining allowance and investigated probable casting-model and setup causes of geometric nonconformities, proposing corrections.

Cam-Driven Capping Machine **2025–26**
Arts et Métiers — Mechanical Design and Optimization
- **Co-optimized a cam-driven capping mechanism across 26,691 design combinations**, reducing calculated maximum pressure angle **10.75%, from 32.92° to 29.38°**, and meeting a 30° target.
- Co-developed motion-law, kinematic and contact calculations for a **45 mm stroke and 2.0 s cycle**; delivered cam CAD and an assembly drawing while checking curvature, follower contact and Hertz pressure constraints.

LEADERSHIP, VOLUNTEERING AND STUDENT ENGAGEMENT

Vice-President & Co-Founder — Student Wellbeing Association **2026**
Arts et Métiers
- **Co-founded a student wellbeing association** to address needs identified through student surveys and conversations; helped establish an **on-campus relaxation room that opened in April 2026**.

Volunteer Tutor — Middle and High School Students **Sept 2022 – Present**
- Provide regular volunteer tutoring in **mathematics, physics and engineering sciences** to students from underserved backgrounds; adapt explanations and study methods to develop structured reasoning, confidence and learning autonomy.

Grand Défi — School Renovation Volunteer **2024**
- Helped renovate a primary school and organize activities for children with fellow students, contributing to repainting, decoration and community-garden preparation.

SKILLS AND INTERESTS

Programming & ML: Python (data analysis, numerical computation, scientific optimization), PyTorch (current coursework), MATLAB, Linux (current coursework), LaTeX
Robotics & Vision: Multi-robot hardware integration, ArUco localization, camera calibration, real-time trajectory testing, ROS 2 (current coursework)
Modeling & Simulation: CFD (STAR-CCM+), FEA (Abaqus), CAD (Fusion 360, 3DEXPERIENCE), Simulink, digital twins
Languages: French (native), English (TOEFL iBT: 4.5/6), Italian (basic), Spanish (basic)
Interests: Robot learning, physical AI, applied reinforcement learning, basketball, strength training